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Criteria C and D_SA 2
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Summative Summative 0/18 Students Thursday at 10:05 AM
IB MYP Design - Ms Ishita (MYP2) B

Task Details

SA Template

ATL Skills

Thinking Skills – Critical Thinking

- Evaluate evidence, arguments, and existing products to justify a proposed design solution.
- Identify strengths, limitations, and opportunities for improvement based on testing evidence.

Research Skills – Information Literacy

- Collect, record, and verify data from primary and secondary sources.
- Interpret testing results to evaluate the effectiveness of the designed solution.

Self-Management Skills – Organisation

- Plan and manage the steps required to create the solution.
- Organise technical evidence, document changes, and complete the ePortfolio systematically.

Concepts

Key Concept: Communities

Related Concepts: Function, Innovation

Content

Planning, creating, testing, and evaluating a sensor-based automated system using a sensor, Arduino microcontroller, and output component. Students will develop their chosen design into a simulation using Wokwi, demonstrate technical skills, document changes made during creation, test the system against its design specifications, and propose evidence-based improvements.

Statement of Inquiry

Innovative systems function effectively when engineers use researched, tested evidence to sense a community's needs and design a justified, responsive solution.

Task Details

Situation

You are a student design engineer developing a sensor-based automated system to address a real community problem. Having

investigated the problem, analysed existing products, developed design specifications, and selected a final design, you must now plan and create your proposed solution.

Your task is to translate your chosen design into a working simulation using Wokwi. The system must use an appropriate sensor, an Arduino microcontroller, and an output component to detect a condition and trigger a suitable response.

Throughout the creation process, you must demonstrate your technical skills, document the steps involved in developing the solution, and record any changes made to your original design or plan. You must also provide evidence of your circuit construction, Arduino code, and debugging or modification process.

Once your solution has been created, you will test its functionality and collect evidence of its performance. You will evaluate how effectively the system meets your design specifications and addresses the identified community problem. Based on your findings, you will identify its strengths and limitations and propose justified modifications for future improvement.

You will document your work in the **MYP Design ePortfolio: Criteria C & D – Creating & Evaluating the Solution**. Your submission must demonstrate a clear connection between your chosen design, the solution you create, the evidence collected through testing, and your evaluation.

Role

A student design engineer responsible for planning, creating, testing, and evaluating a sensor-based automated solution for a real community problem.

Audience

The intended users of the proposed solution, members of the affected community, and the design panel evaluating the functionality, effectiveness, and potential for improvement of the solution.

Product

A completed **MYP Design ePortfolio covering Criteria C and D**, containing the following deliverables.

Criterion C: Creating the Solution

1. **Strand i – Planning the Solution:** A logical plan showing the sequence of steps required to create the solution.
2. **Strand ii – Technical Skills:** Screenshots or other relevant evidence of technical skills demonstrated during the creation process, accompanied by reflections on the skills used.
3. **Strand iii – Creating the Solution:** A Wokwi circuit simulation, Arduino code, and a reflection explaining the creation process and coding logic.
4. **Strand iv – Changes During Creation:** A record of changes made during creation, the reasons for those changes, and reflections on their impact on the solution.

Criterion D: Evaluating the Solution

1. **Strand i – Evaluation:** Evidence of the completed solution and its final state.
2. **Strand ii – Testing:** Evidence of testing the solution, including relevant screenshots and observations.
3. **Strand iii – Evaluating Success:** An evaluation identifying the strengths and limitations of the solution in relation to its intended purpose and design specifications.
4. **Strand iv – Future Improvements:** A proposed redesign or modification, supported by a sketch or diagram and a

reasoned explanation of how it could improve the solution.

The ePortfolio must also include the final reflection and completed submission checklist provided in the template.

Standards

Your work will be assessed against **MYP Design Criterion C: Creating the Solution and Criterion D: Evaluating**, using the achievement-level descriptors and Task-Specific Clarifications (TSC) provided below.

Task-Specific Clarifications (TSC)Criterion C: Creating the Solution

Grade boundary	IB rubric	Task-Specific Clarification (TSC)
0	The student does not reach a standard described by any of the descriptors below.	The student does not provide sufficient evidence to meet any of the task-specific expectations below.
1–2	i. Demonstrates minimal technical skills when making the solution. ii. Creates the solution, which functions poorly and is presented in an incomplete form.	i. Provides an incomplete or unclear plan for creating the sensor-based system. ii. Provides limited screenshots or evidence of technical skills. iii. Presents an incomplete Wokwi circuit or Arduino code that functions poorly. iv. Records few or no changes made during creation, with little explanation.
3–4	i. Outlines each step in a plan that contains some details, resulting in peers having difficulty following the plan to create the solution. ii. Demonstrates satisfactory technical skills when making the solution. iii. Creates the solution, which partially functions and is adequately presented. iv. Outlines changes made to the chosen design or plan when making the solution.	i. Outlines the main steps for creating the sensor-based system, but some steps, resources, or sequencing details are unclear. ii. Demonstrates satisfactory technical skills and provides some screenshots of the creation process. iii. Presents a Wokwi circuit and Arduino code that demonstrate some intended functions, although the system may not work consistently. iv. Identifies changes made to the design or plan and outlines basic reasons for them.
5–6	i. Constructs a plan that considers time and resources, sufficient for peers to be able to follow to create the solution. ii. Demonstrates competent technical skills when making the solution. iii. Creates the solution, which functions as intended and is presented appropriately. iv. Outlines changes made to the chosen design and plan when making the solution.	i. Constructs a logical plan that includes the main creation steps, sequence, time, and resources needed to develop the sensor-based system. ii. Demonstrates competent technical skills in Wokwi circuit construction and Arduino coding, supported by relevant screenshots and reflections. iii. Presents a functional simulation with the sensor, Arduino, and output component connected and programmed to perform the intended task. iv. Records changes made during creation and outlines their reasons.
7–8	i. Constructs a logical plan, which outlines the efficient use of time and resources, sufficient for peers to be able to follow to create the solution. ii. Demonstrates excellent technical skills when making the solution. iii. Follows the plan to create the solution, which functions as intended and is presented appropriately. iv. Explains	i. Constructs a detailed, logically sequenced plan that clearly communicates the efficient use of time and resources, enabling a peer to recreate the system. ii. Demonstrates excellent technical skills through accurate circuit construction, effective Arduino coding, debugging, and clear evidence of the creation process. iii. Follows the plan to produce a fully functional Wokwi simulation that accurately implements the chosen design and is clearly presented. iv. Explains changes made to the design or plan and justifies the

Grade boundary	IB rubric	Task-Specific Clarification (TSC)
	changes made to the chosen design and plan when making the solution.	reasons for those changes in relation to functionality and the intended purpose.

Criterion D: Evaluating

Grade boundary	IB rubric	Task-Specific Clarification (TSC)
0	The student does not reach a standard described by any of the descriptors below.	The student does not provide sufficient evidence to meet any of the task-specific expectations below.
1–2	i. Describes a testing method, which is used to measure the success of the solution. ii. States the success of the solution.	i. Describes a basic test for the sensor-based system, with limited details about how it will be conducted. ii. States whether the system works, with little or no supporting evidence from testing or reference to the design specifications.
3–4	i. Describes a relevant testing method, which generates data, to measure the success of the solution. ii. Outlines the success of the solution against the design specification based on relevant product testing. iii. Lists the ways in which the solution could be improved. iv. Outlines the impact of the solution on the client/ target audience.	i. Describes a relevant test of the sensor, programmed response, or output and records some observations or results. ii. Outlines whether the system meets some design specifications, using relevant testing evidence. iii. Lists possible improvements to the circuit, code, or system response. iv. Outlines how the proposed system could benefit the intended community or users.
5–6	i. Describes relevant testing methods, which generate data, to measure the success of the solution. ii. Describes the success of the solution against the design specification based on relevant product testing. iii. Outlines how the solution could be improved. iv. Describes the impact of the solution on the client/target audience, with guidance.	i. Describes relevant tests for the sensor-based system and records results that demonstrate how the system responds under the tested conditions. ii. Describes the extent to which the system meets the design specifications, referring to testing results and relevant evidence. iii. Outlines specific improvements to address identified limitations in the circuit, code, or system response. iv. Describes the likely impact of the proposed system on the intended users or community, with guidance.
7–8	i. Describes detailed and relevant testing methods, which generate accurate data, to measure the success of the solution. ii. Explains the success of the solution against the design specification based on authentic product testing. iii.	i. Describes detailed, relevant testing methods and records accurate results across appropriate test conditions to measure the system's performance. ii. Explains the success of the system against the design specifications, explicitly linking authentic simulation or product-testing evidence to each relevant specification. iii. Describes specific, feasible improvements that address identified weaknesses and explains how they could improve functionality. iv. Describes the potential impact of the

Grade boundary	IB rubric	Task-Specific Clarification (TSC)
	Describes how the solution could be improved. iv. Describes the impact of the solution on the client/target audience.	system on the intended users or community, linking the explanation to the original problem and the solution's demonstrated performance.

Submission Requirements

- Complete all sections of the provided MYP Design ePortfolio.
- Include screenshots of the Wokwi circuit, Arduino code, technical skills, and testing process wherever indicated in the template.
- Ensure that the reflections explain your decisions and are supported by relevant evidence.
- Refer to the design specifications developed in Criteria A and B when evaluating the solution.
- Document changes made during creation, including the reason for each change and its effect on the solution.
- Include testing observations and evidence that demonstrate the performance of the sensor-based system.
- Propose realistic future improvements supported by a redesign sketch or diagram.
- Complete the final reflection and submission checklist before submitting the ePortfolio on ManageBac.

Assessment Criteria

C: Creating the solution

Achievement Level	Descriptor
0	The student does not reach a standard described by any of the descriptors below.
1 - 2	The student: i. demonstrates minimal technical skills when making the solution ii. creates the solution, which functions poorly and is presented in an incomplete form.
3 - 4	The student: i. outlines each step in a plan that contains some details, resulting in peers having difficulty following the plan to create the solution ii. demonstrates satisfactory technical skills when making the solution iii. creates the solution, which partially functions and is adequately presented iv. outlines changes made to the chosen design or plan when making the solution.

Achievement Level	Descriptor
5 - 6	The student: i. constructs a plan, which considers time and resources, sufficient for peers to be able to follow to create the solution ii. demonstrates competent technical skills when making the solution iii. creates the solution, which functions as intended and is presented appropriately iv. outlines changes made to the chosen design and plan when making the solution.
7 - 8	The student: i. constructs a logical plan, which outlines the efficient use of time and resources, sufficient for peers to be able to follow to create the solution ii. demonstrates excellent technical skills when making the solution iii. follows the plan to create the solution, which functions as intended and is presented appropriately iv. explains changes made to the chosen design and plan when making the solution.

D: Evaluating

Achievement Level	Descriptor
0	The student does not reach a standard described by any of the descriptors below.
1 - 2	The student: i. describes a testing method, which is used to measure the success of the solution ii. states the success of the solution.
3 - 4	The student: i. describes a relevant testing method, which generates data, to measure the success of the solution ii. outlines the success of the solution against the design specification based on relevant product testing iii. lists the ways in which the solution could be improved iv. outlines the impact of the solution on the client/target audience.
5 - 6	The student: i. describes relevant testing methods, which generate data, to measure the success of the solution ii. describes the success of the solution against the design specification based on relevant product testing iii. outlines how the solution could be improved iv. describes the impact of the solution on the client/target audience, with guidance.
7 - 8	The student: i. describes detailed and relevant testing methods, which generate accurate data, to measure the success of the solution ii. explains the success of the solution against the design specification based on authentic product testing iii. describes how the solution could be improved iv. describes the impact of the solution on the client/target audience.